



Inspiring Excellence

CSE 251

Electronic Devices and Circuits

Lecture 3

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Outline

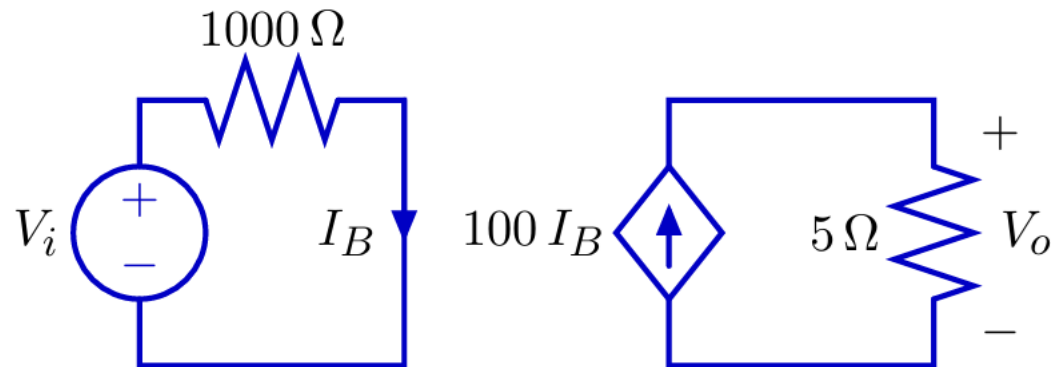
- **Operational Amplifier: Introduction**
 - Dependent Sources
 - Op-Amp: Circuit Symbols and terminal
 - Op-Amp: VTC (Voltage Transfer Characteristics)
 - Linear Amplification
 - (Positive and Negative) Saturation
 - Op-Amp: Examples
 - Op-Amp: Physical Entity

Dependent Sources

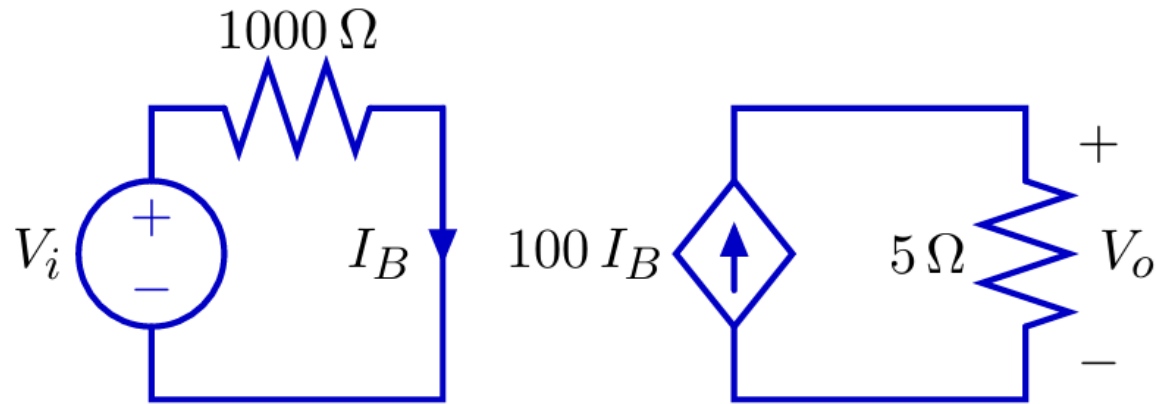
To analyze op-amps, we must understand **dependent source**.

A dependent source generates a voltage or current whose value depends on another voltage or current.

Example: current-controlled current source



Dependent Sources



$$I_B = \frac{V_i}{1000\ \Omega}$$

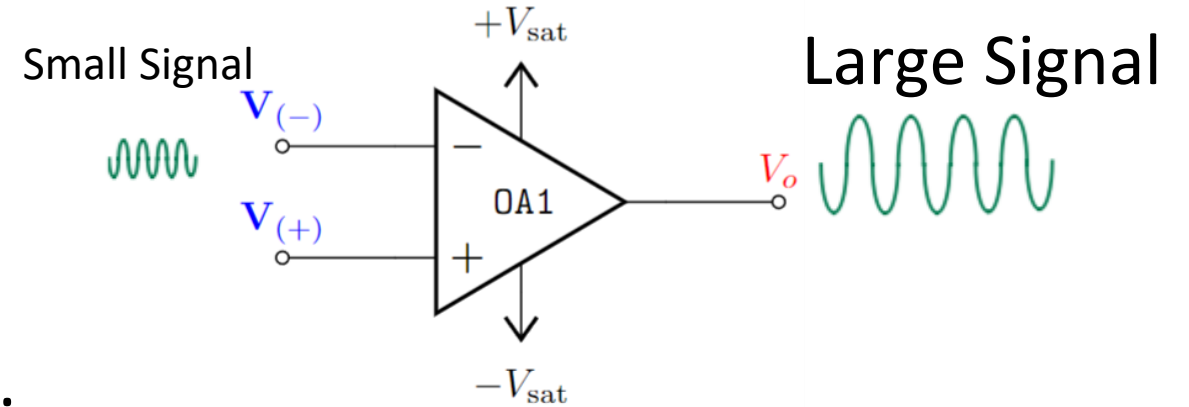
$$V_o = 100 I_B \times 5\ \Omega$$

$$\text{Voltage Gain: } \frac{V_o}{V_i} = \frac{1}{2}$$

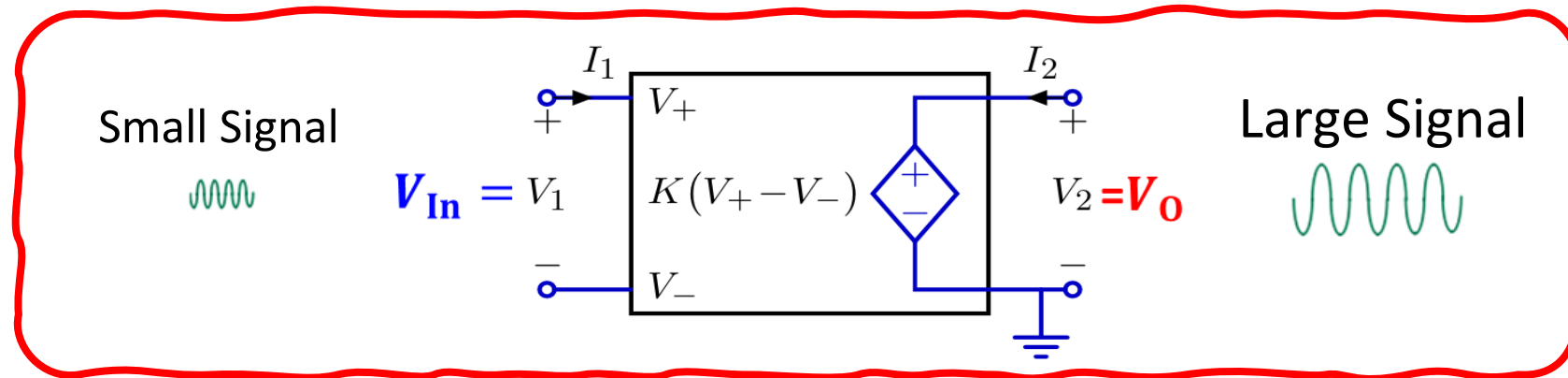
$$= 100 \frac{V_i}{1000\ \Omega} \times 5\ \Omega = \frac{1}{2} V_i$$

Operational Amplifier: Introduction

- **Operational:**
Mathematical Operations
- **Amplifier:**
Amplifies input signal/voltage.

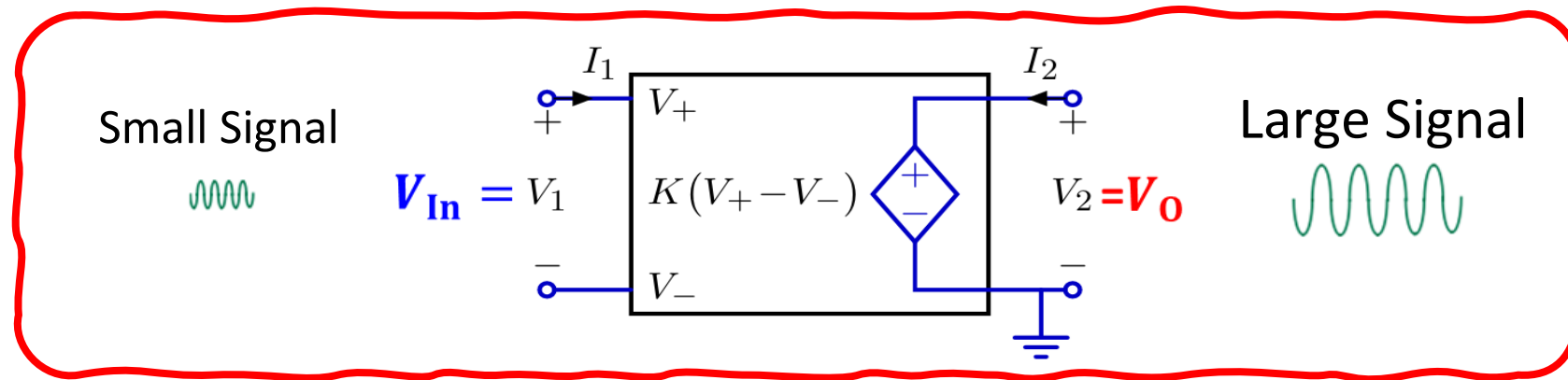


An **op-amp (operational amplifier)** can be modelled by a **voltage- controlled voltage source**.



Operational Amplifier: Introduction

An **op-amp (operational amplifier)** can be represented by a **voltage- controlled voltage source**.

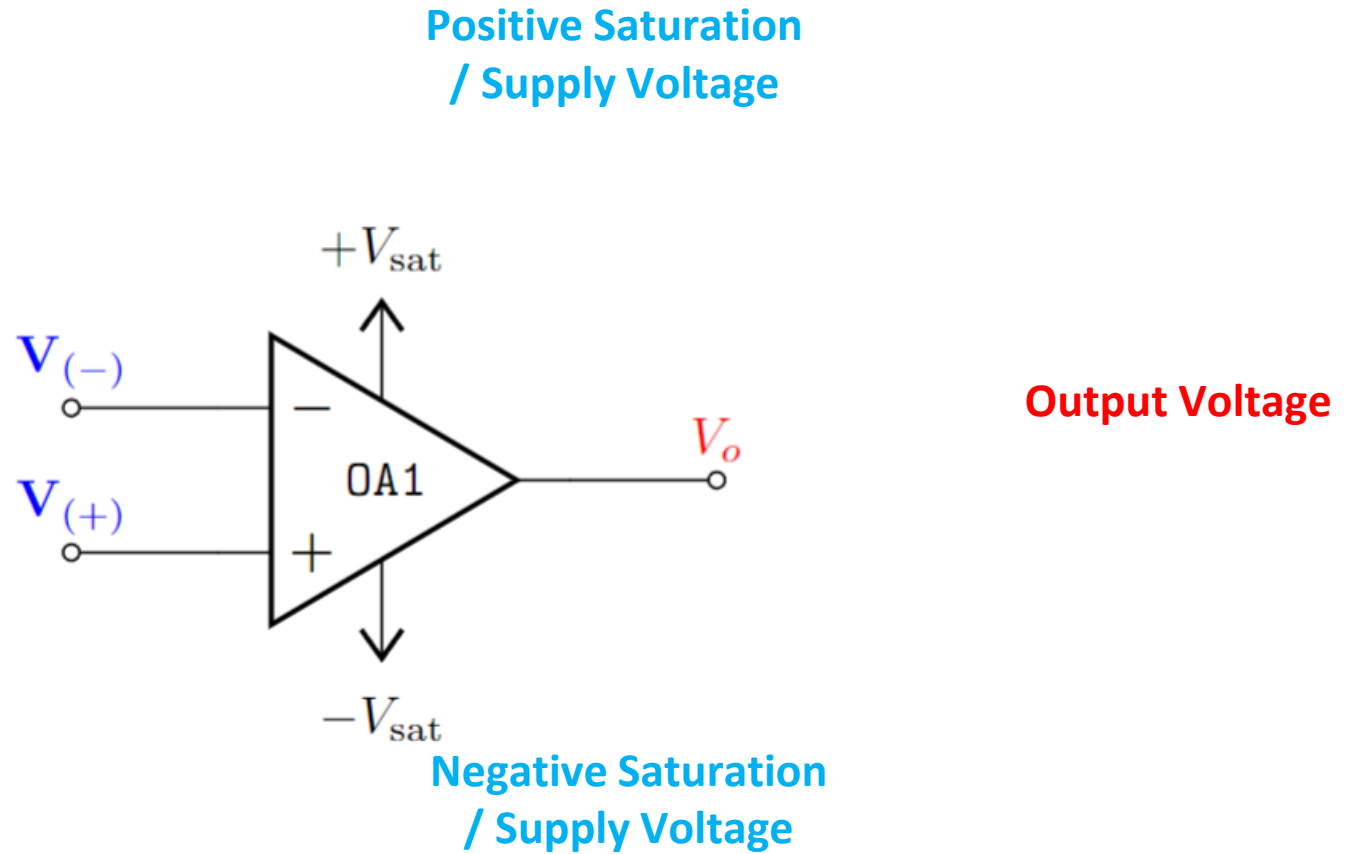


$I_1 = 0$ and $V_O = K V_{In}$ where K is large (typically $K > 10^5$).

Op-Amp: Circuit Symbols and terminal

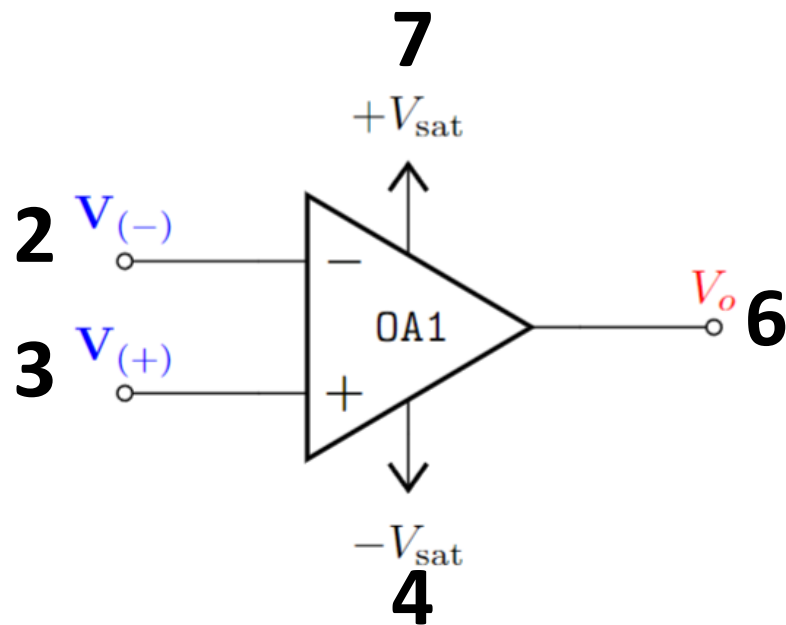
Inverting
terminal voltage

Non-inverting
terminal voltage



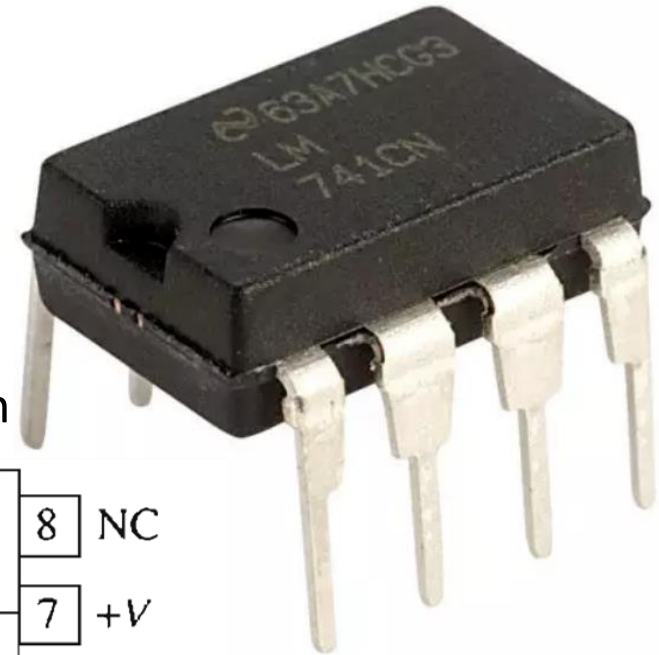
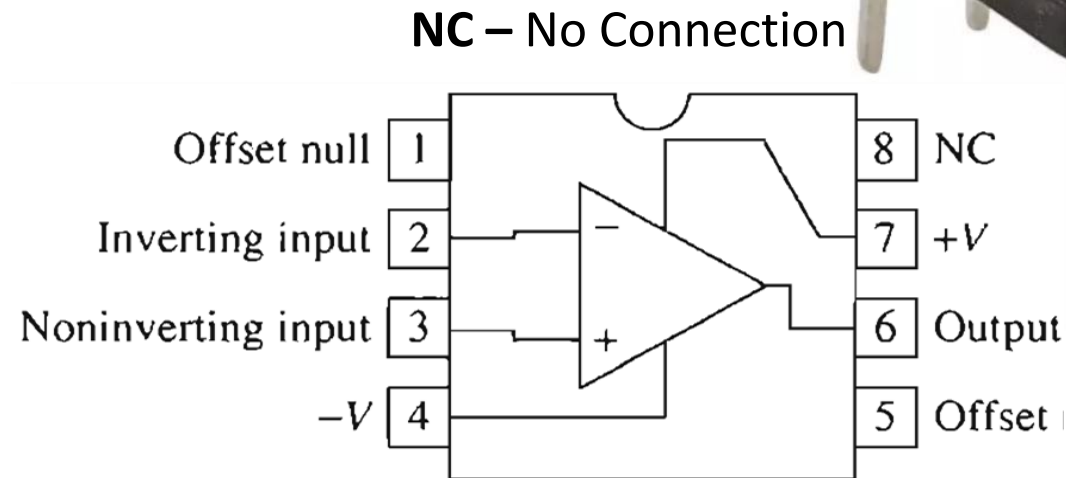
Op-Amp: Physical Entity

Difference Amplifier – Amplifies the voltage difference between two terminals.



Circuit symbol for the general-purpose op amp.

Pin numbering is that for an **8-pin mini-DIP package**



μ A-741C

Manufacturer ID

Part Identification Number (PIN)

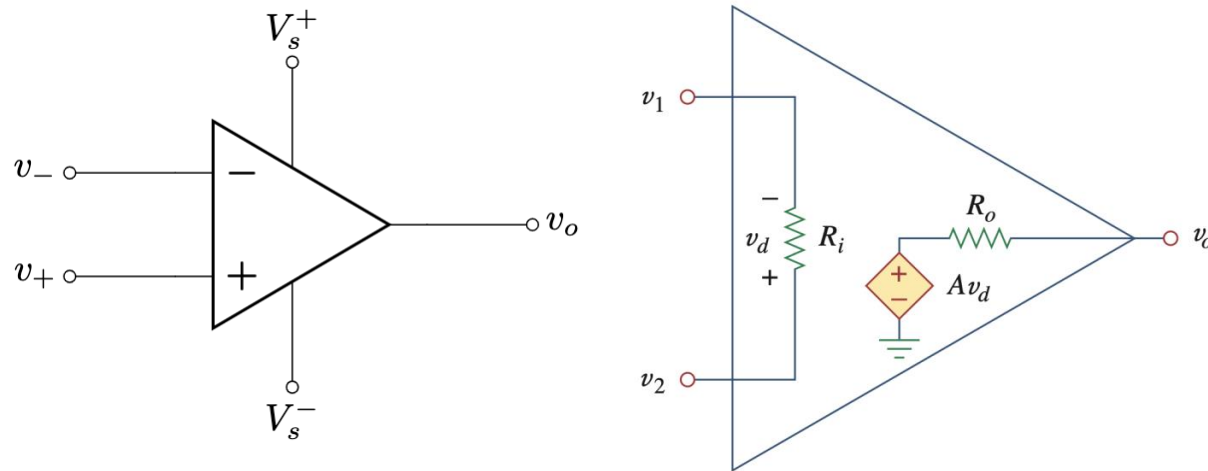
C: Commercial ($0^{\circ} - 70^{\circ}\text{C}$)
I: Industrial ($-25^{\circ} - 85^{\circ}\text{C}$)
M: Military ($-55^{\circ} - 125^{\circ}\text{C}$)

Operational Amplifiers

- An operational amplifier, or **op-amp** for short, is a versatile and powerful integrated circuit that is widely used in a variety of electronic applications.
- An Op-Amp is designed so that it performs some mathematical operations when external components, such as resistors and capacitors, are connected to its terminals.
- The op amp is an electronic device consisting of a complex arrangement of resistors, transistors, capacitors, and diodes. A full discussion of what is inside the op amp is beyond the scope of this course. For now, it will suffice to treat the op amp as a circuit building block and simply study what takes place at its terminals.

Equivalent Circuit

Since op amp is an amplifier, the internal circuit can be modeled using a **voltage controlled voltage source** (VCVS)! (actual circuit is complicated)



$v_1 = v_-$ = voltage of inverting terminal
 $v_2 = v_+$ = voltage of noninverting terminal

$v_d = v_+ - v_- = v_2 - v_1$
= differential input voltage for VCVS

A = Open loop gain

R_i = Input resistance

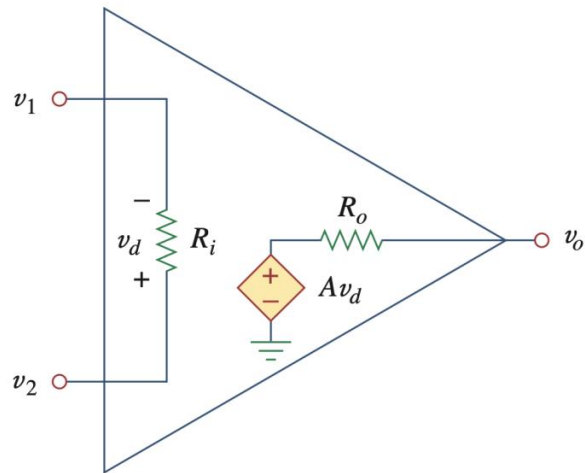
R_o = Output resistance

The op amp senses the difference between the two inputs, multiplies it by the gain A , and causes the resulting voltage to appear at the output. Thus, the output v_o is given by

$$v_o = Av_d = A(v_2 - v_1) = A(v_+ - v_-)$$

Equivalent Circuit

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Parameter	Typical Range

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Op-Amp: VTC – Linear Amplification

Voltage Transfer Characteristics (VTC)

- Shows how the **output voltage** varies with the **input voltage** $V_O(V_d)$

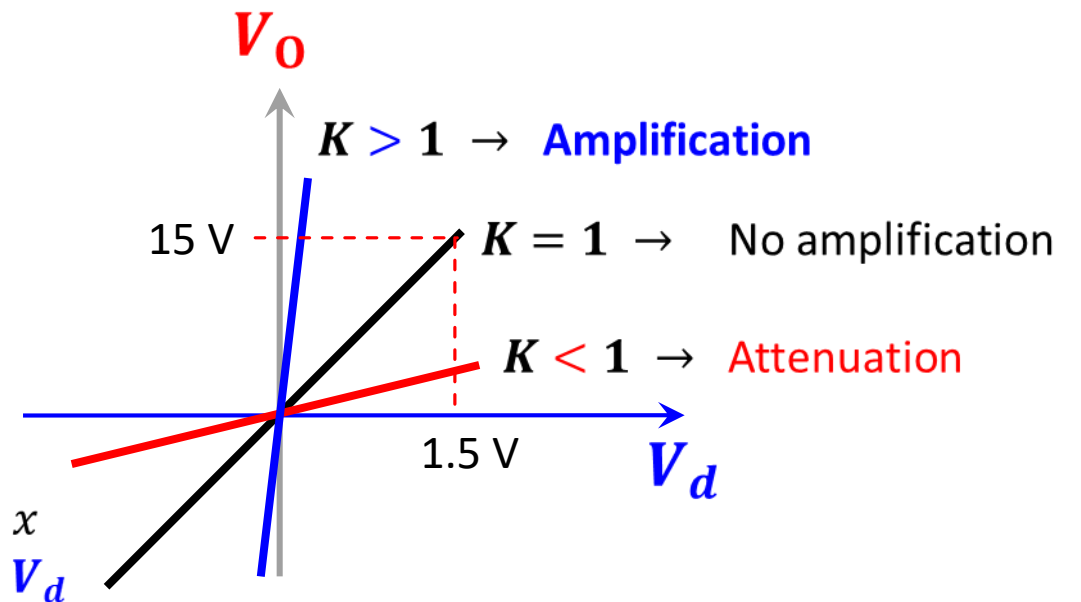
- y – axis $\rightarrow V_O$

- x – axis $\rightarrow V_d$

- Slope: $K = \frac{\Delta V_O}{\Delta V_d} = \frac{V_O}{V_d}$

If the line passes through origin:

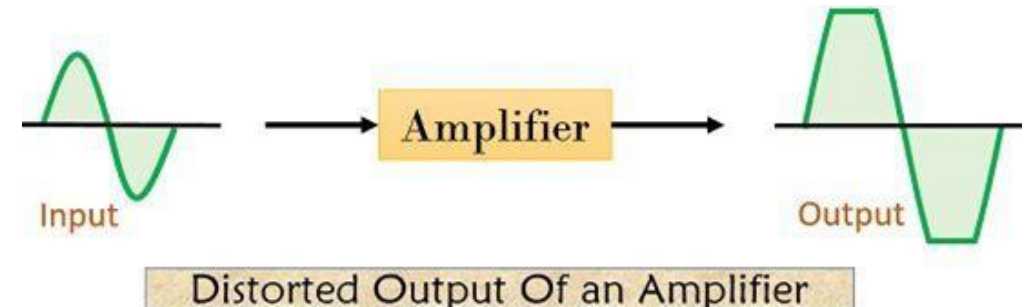
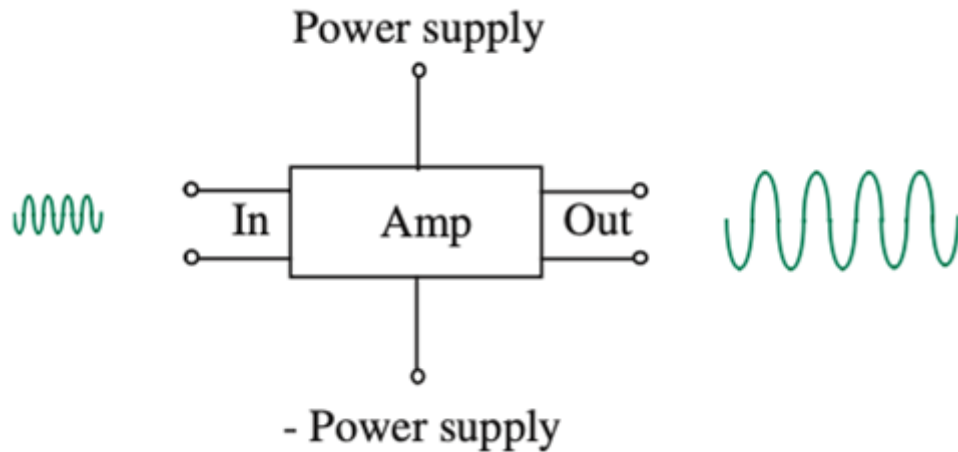
$$y = K \cdot x$$
$$V_O = K \cdot V_d$$



LINEAR AMPLIFICATION

Op-Amp: VTC – Linear Amplification

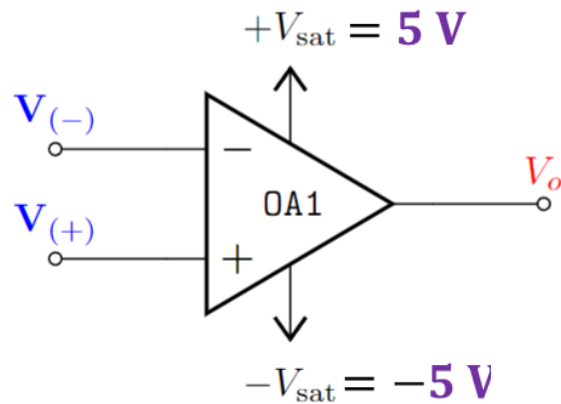
- **Linear Amplification** only takes place within a [valid input range](#).
- Otherwise output will be distorted - - Saturation



The limiting factor of **linear amplification** is determined by the **power supply** to the amplifier

Op-Amp: VTC - Saturation

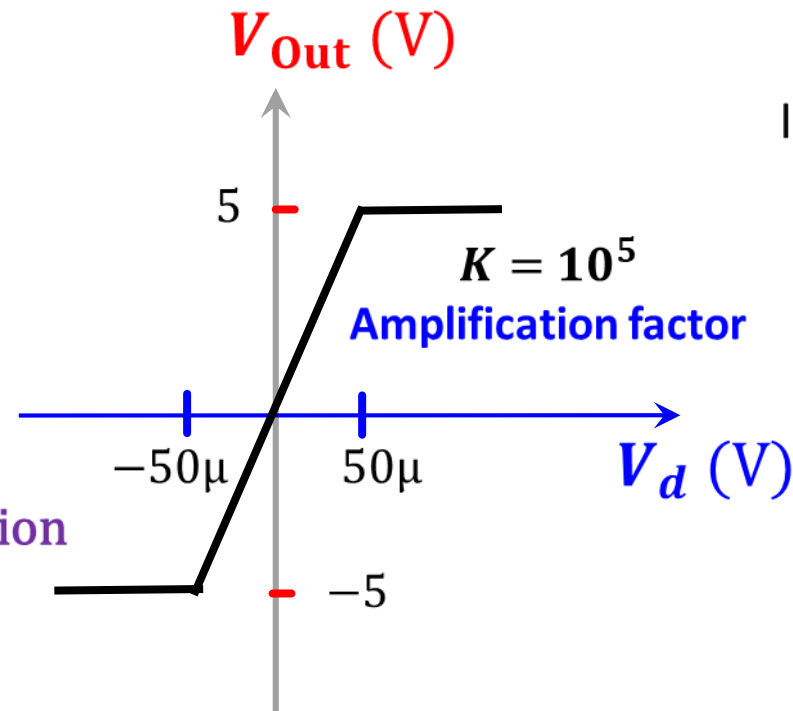
VTC -> Voltage Transfer Characteristics



$$V_o = K \cdot V_{in} : \text{When } -5\text{ V} < V_o < 5\text{ V}$$

$K \rightarrow 10^5$: Gain / Amplification

Non-Linear characteristics



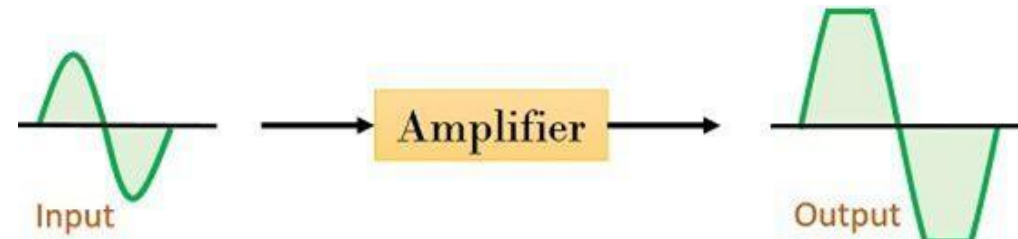
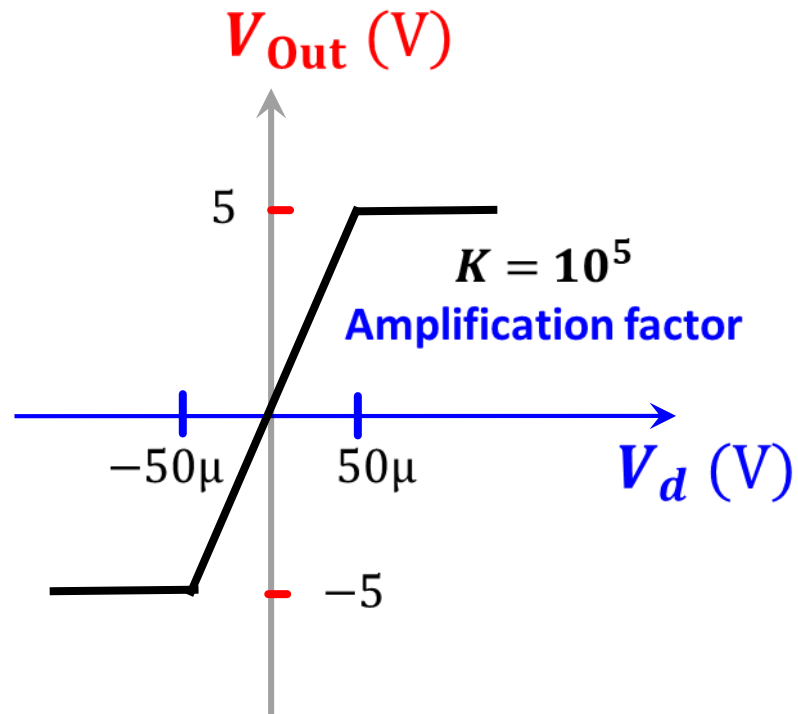
If $V_d > 50\mu$: Positive Saturation

$$\Rightarrow V_o = 5$$

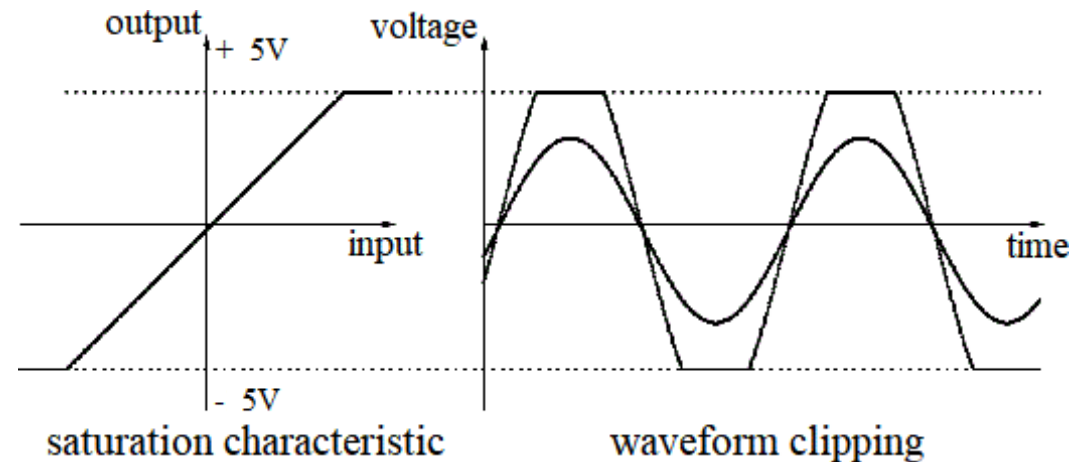
If $V_d < -50\mu$: Negative Saturation

$$\Rightarrow V_o = -5$$

Op-Amp: VTC - Saturation

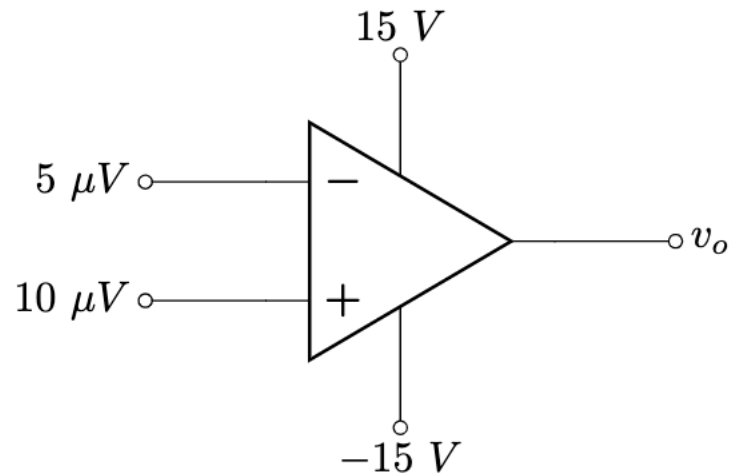


Distorted Output Of an Amplifier

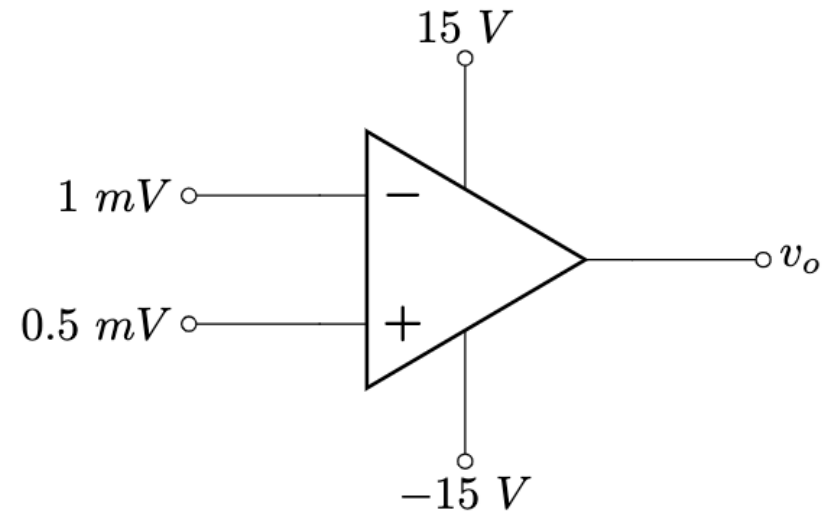


Op-Amp: Examples

Find v_o



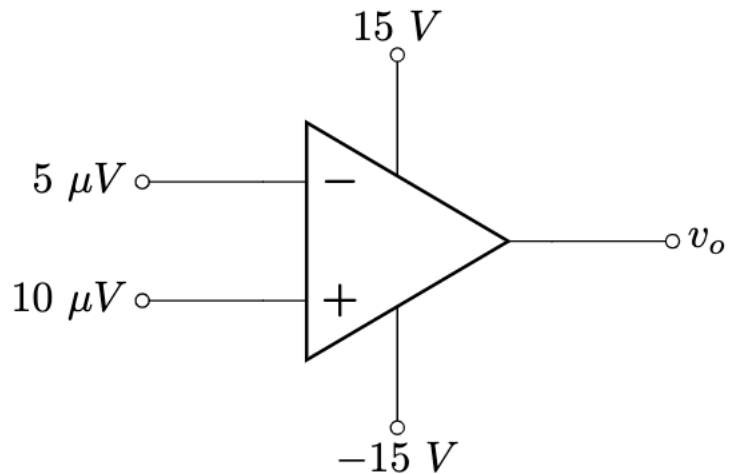
$$A = 2 \times 10^5$$



$$A = 2 \times 10^5$$

Example 1

- Find v_o



$$A = 2 \times 10^5$$

Solution:

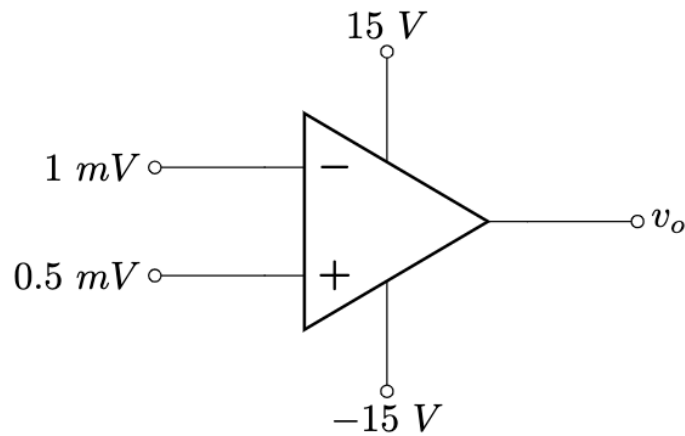
$$v_d = v_+ - v_- = 10\ \mu V - 5\ \mu V = 5\ \mu V$$

$$\Rightarrow v_o = Av_d = (2 \times 10^5) \times (5 \times 10^{-6}) = 1\ V$$

Since this is in between $-V_{CC}$ and V_{CC} , output valid

Example 2

- Find v_o



$$A = 2 \times 10^5$$

Solution:

$$v_d = v^+ - v^- = 0.5 \text{ mV} - 1 \text{ mV} = -0.5 \text{ mV}$$

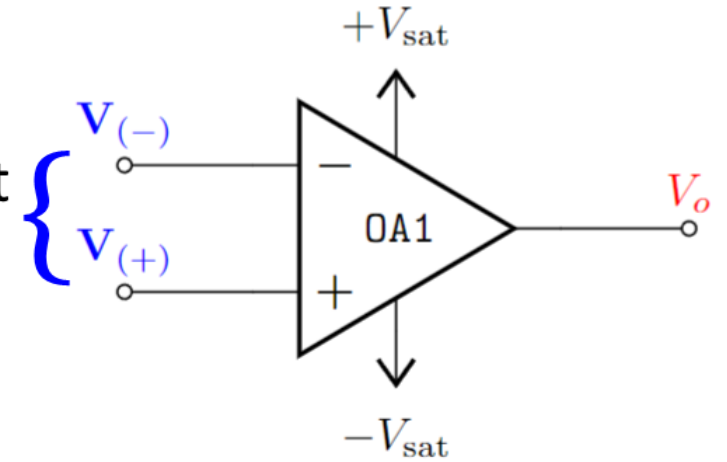
$$\rightarrow v_o = Av_d = (2 \times 10^5) \times (-0.5 \times 10^{-3}) = -100 \text{ V}$$

However, the output must be limited within the range of $-V_{CC}$ to $+V_{CC}$. Therefore, the highest output voltage can be -15V.

$$\therefore v_o = -15 \text{ V}$$

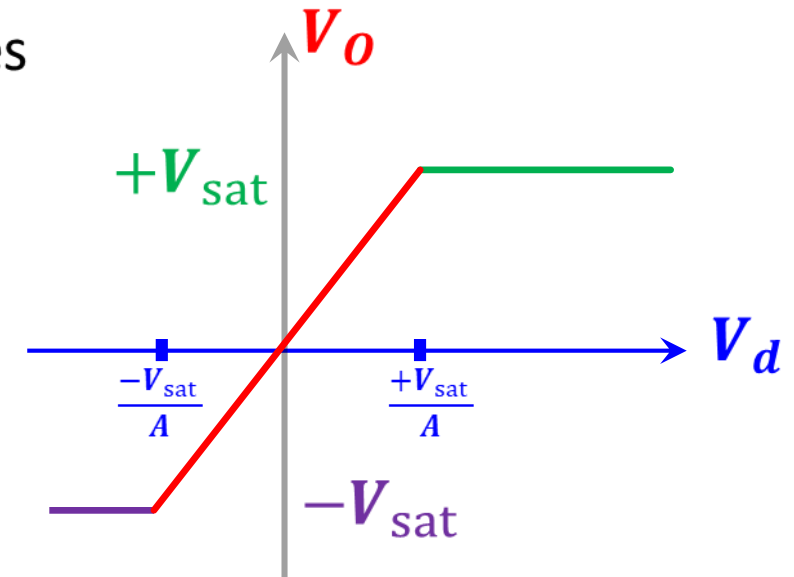
Op-Amp: Summary

Op-Amp **Amplifies** the difference between the voltages at its two input terminals - V_d



However, the **Amplification** is limited within voltage levels defined by the positive and negative saturation voltages $[-V_{sat}, +V_{sat}]$.

The “ideal” op-amp behaves like a **voltage dependent voltage source** within the linear region.

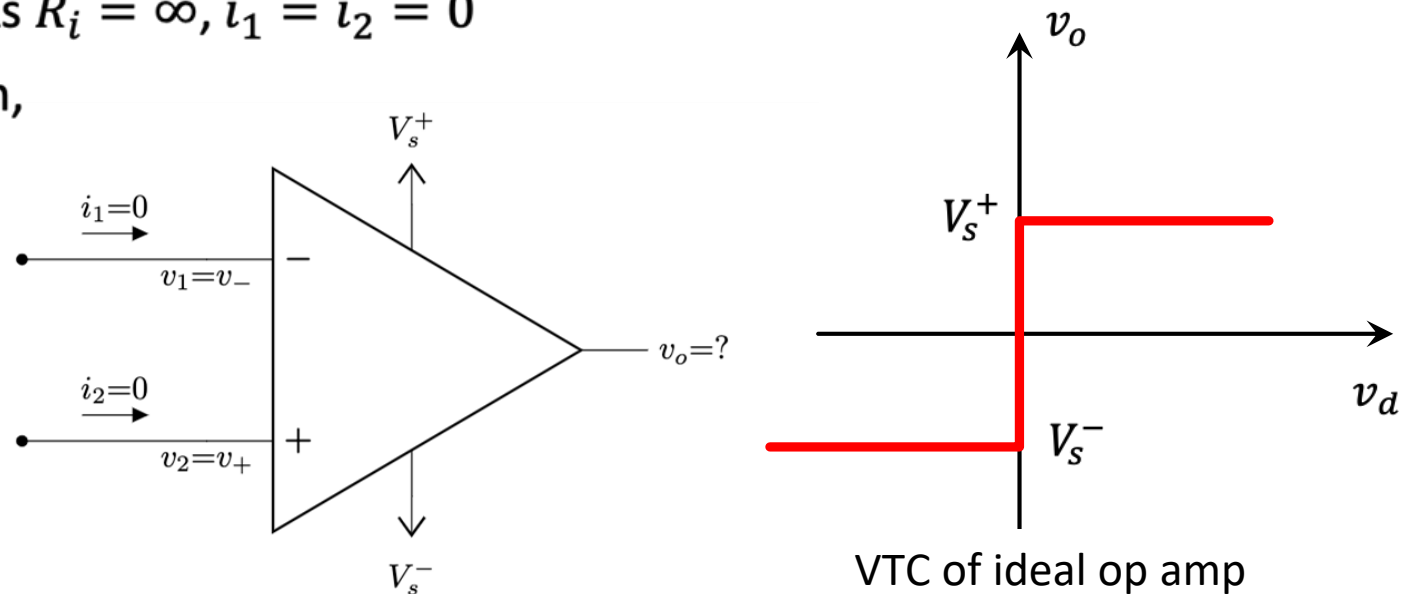


The Ideal Op-Amp

- To facilitate the understanding of op amp circuits, we will assume **ideal op amps**
 - Infinite open-loop gain, $A = \infty$
 - Infinite input resistance, $R_i = \infty = \text{open circuit}$
 - Zero output resistance, $R_o = 0 = \text{short circuit}$
- Although an ideal op amp provides only an approximate analysis, most modern amplifiers have such large gains and input impedances that the approximate analysis is a good one.
- **Circuit solving become much simpler.** As $R_i = \infty, i_1 = i_2 = 0$
- Since $A = \infty$, in open-loop configuration, v_o will either be positive saturated or negative saturated (why?)

$$v_o = \begin{cases} V_s^+ & \text{if } v_d > 0 \Rightarrow v_2 > v_1 \\ V_s^- & \text{if } v_d < 0 \Rightarrow v_2 < v_1 \end{cases}$$

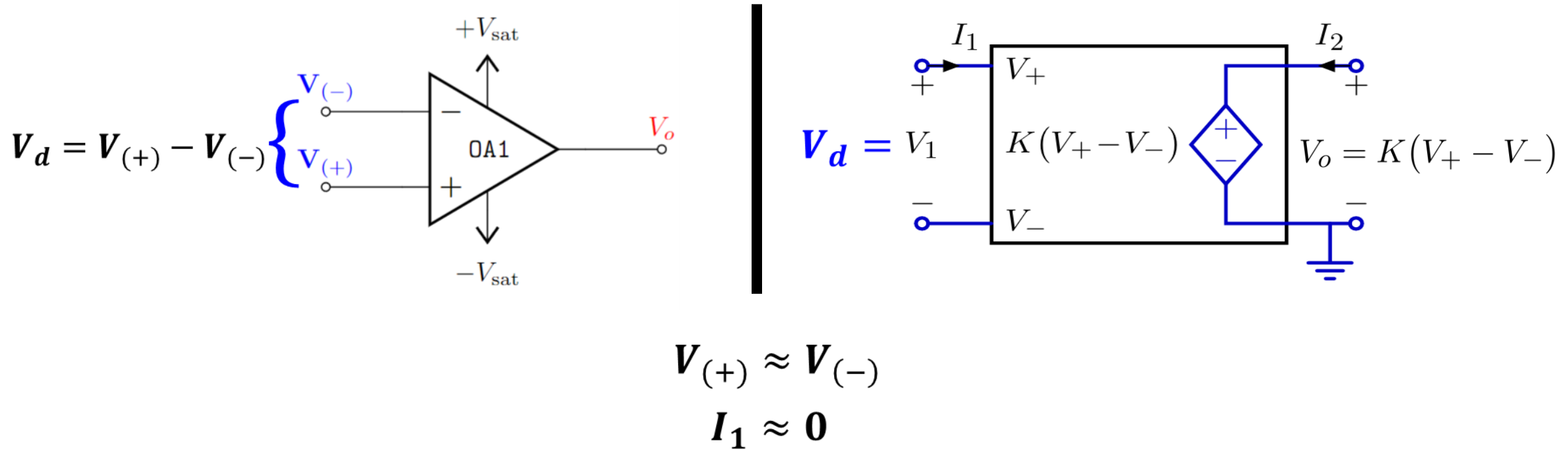
Reminder: $v_d = v_2 - v_1 = v_+ - v_-$



Op-Amp: Circuit Modelling

Voltage controlled voltage Source

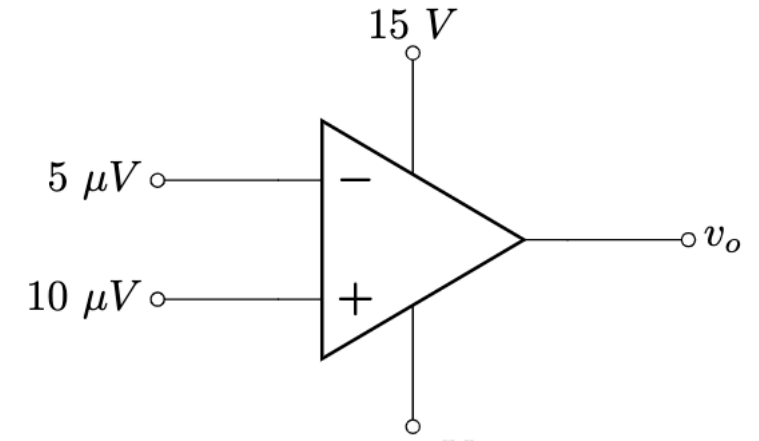
“Ideal” op-amp approximation



Types of Op-Amp configuration

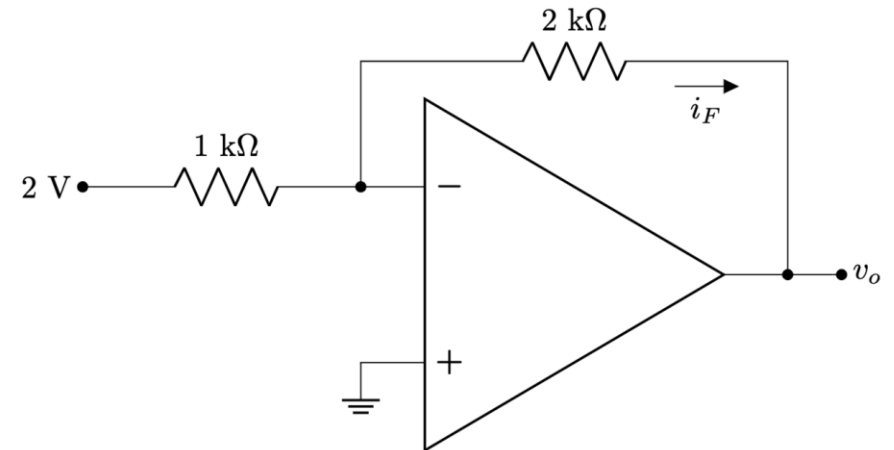
1. Open loop configuration:

No physical connection between input and output



1. Closed loop configuration:

Feedback from output terminal



Application - Comparator

- A comparator compares two voltages to determine which is larger.
- The comparator is essentially an op-amp operated in an open-loop configuration
- Two types –
 - (1) **Non-inverting**: outputs a positive voltage ($V_H = V_S^+$) when input is greater than reference
 - (2) **Inverting**: outputs a negative voltage ($V_L = V_S^-$) when input is greater than reference
- Application – smoke detector, turning AC on/off automatically, etc (next lecture)

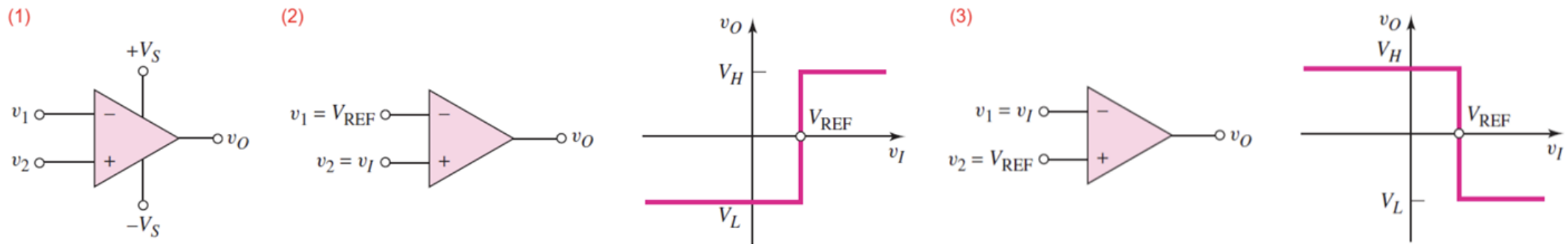


Figure 2: (1) Op-Amp Comparator (2) Noninverting Circuit (3) Inverting Circuit

Comparator Application – Automatic AC

Auto AC ON/OFF

Sensor



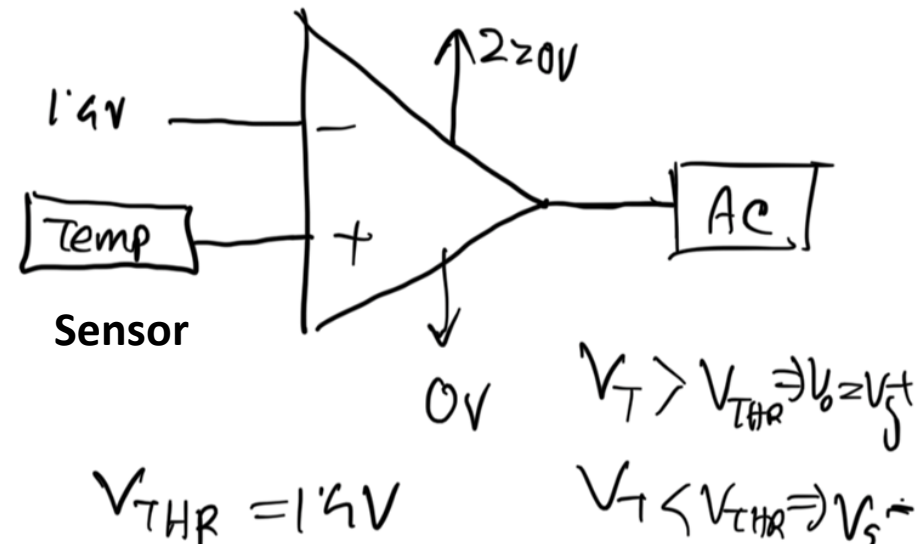
$$V_T \propto T$$

$$23^\circ, V_T = 1.2V$$

$$24^\circ, V_T = 1.4V$$

$$25^\circ, V_T = 1.6V$$

AC should be on if
 $T > 24^\circ C$



Example 1

A valve is used to release (when valve is OPEN,) or maintain (when valve is CLOSED,) water pressure in a water tank. The valve operates on **ACTIVE LOW** logic. (i.e., the valve is OPENED when given a LOW voltage of 1 V, but remains CLOSED when provided a HIGH voltage of 6 V.)

A pressure sensor is installed in the water tank that outputs a voltage linearly proportional to pressure, as shown in the table below.

At 0.5 atm pressure	At 1 atm pressure	At 1.5 atm pressure
$v_{0.5 \text{ atm}} = 0.5 \text{ V}$	$v_{1 \text{ atm}} = 3 \text{ V}$	$v_{1.5 \text{ atm}} = 5.5 \text{ V}$

The pressure in the water tank can be measured by the formula $P = h\rho g$, where P , (in **Pascals (Pa)** unit) is the water pressure, h is the height of water in the tank (in *metres*), $\rho (= 1000 \text{ kgm}^{-3})$ is the density of water and g is the acceleration due to gravity (in ms^{-2}).

[1 atm = 101325 Pa]

- Design** a circuit using Op-Amp comparator to automatically turn OPEN the valve if water level exceeds **10 m**.

Solution:

When $h = 10$ m: $P = h\rho g = 98000$ Pa = 0.967 atm

From the table we can interpolate and find the exact voltage at this pressure.

For $1 - 0.5 = 0.5$ atm pressure change, the voltage changes by 2.5 V

For 1 atm pressure change, the voltage changes by $2.5/0.5$ V = 5 V

So, for $0.967 - 0.5 = 0.467$ atm pressure change, the voltage changes by $5 * 0.467$ V

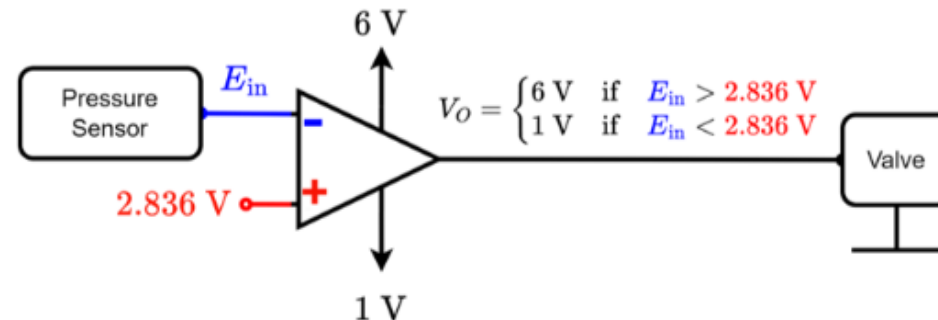
So, voltage at 0.967 atm pressure is $0.5 + 5 * 0.467$ V = 2.836 V

Active low logic:

High water level $\rightarrow$ High pressure $\rightarrow$ **High input voltage** $\rightarrow$ Valve Open -1 V = V_L

Low water level $\rightarrow$ Low pressure $\rightarrow$ **Low input voltage** $\rightarrow$ Valve Closed -6 V = V_H

So the comparator is in inverting configuration. As shown below:

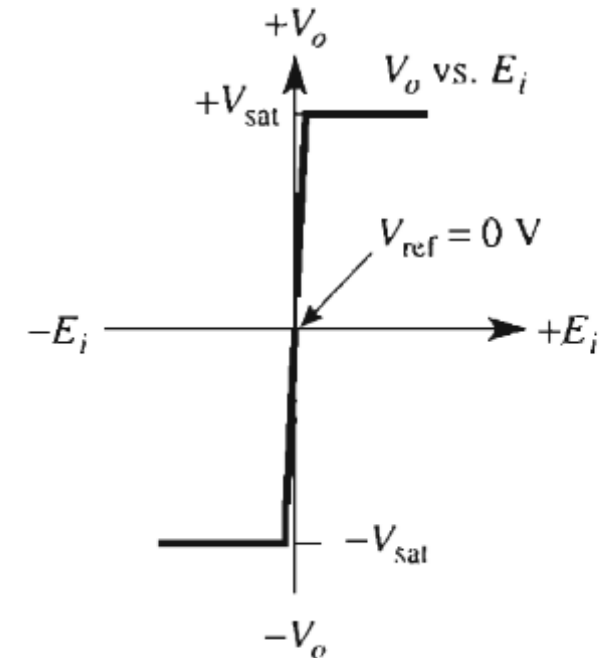
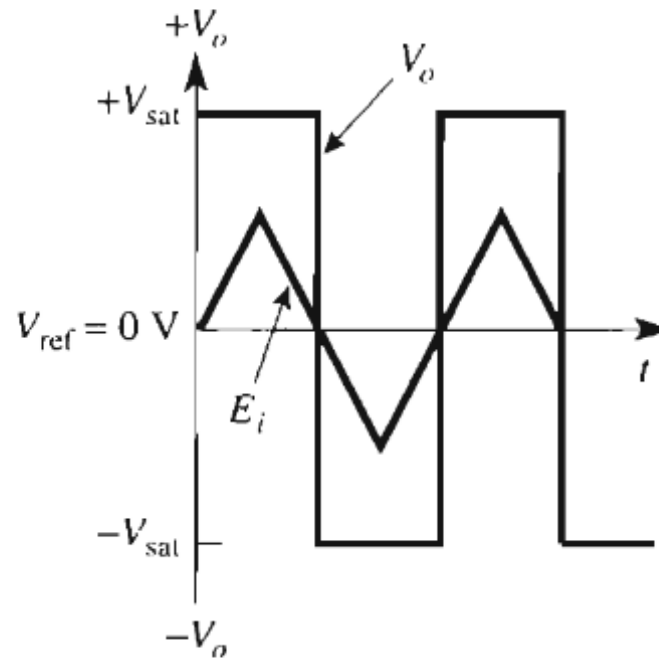
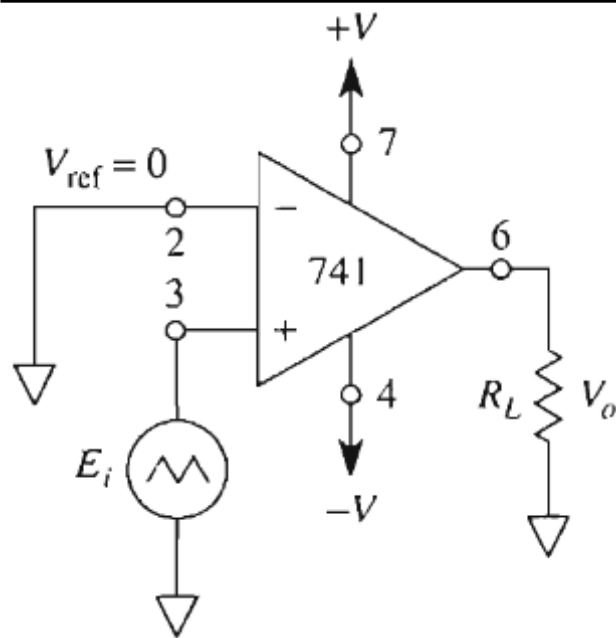


Open Loop Configuration: Comparator

Zero Crossing Detector

Compare values with a reference and pin value to $+V_{sat}$ if voltage is above or to below that.

Non-inverting configuration

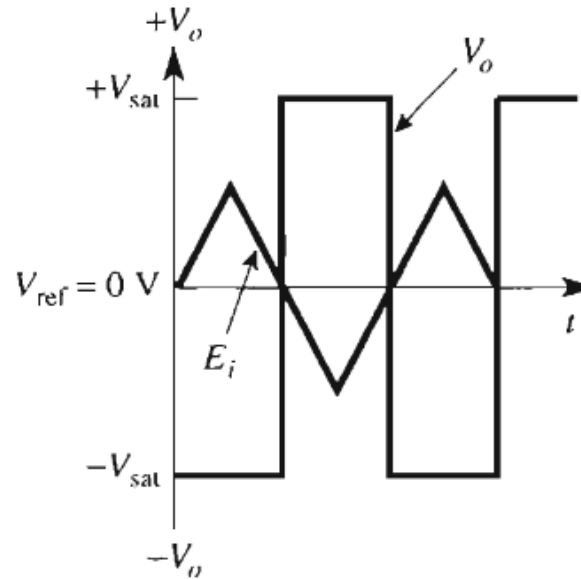
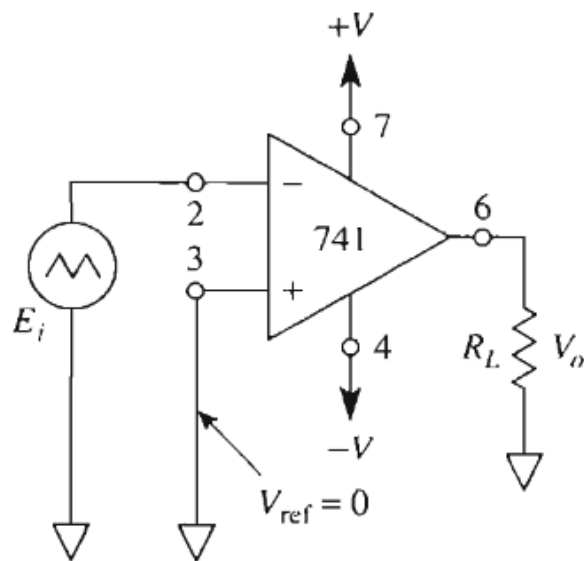


(a) Noninverting: When E_i is above V_{ref} , $V_o = +V_{sat}$.

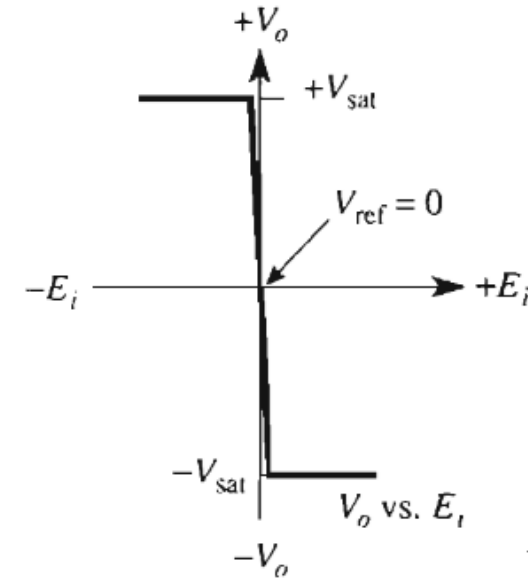
Open Loop Configuration: Comparator

Zero Crossing Detector

Inverting configuration

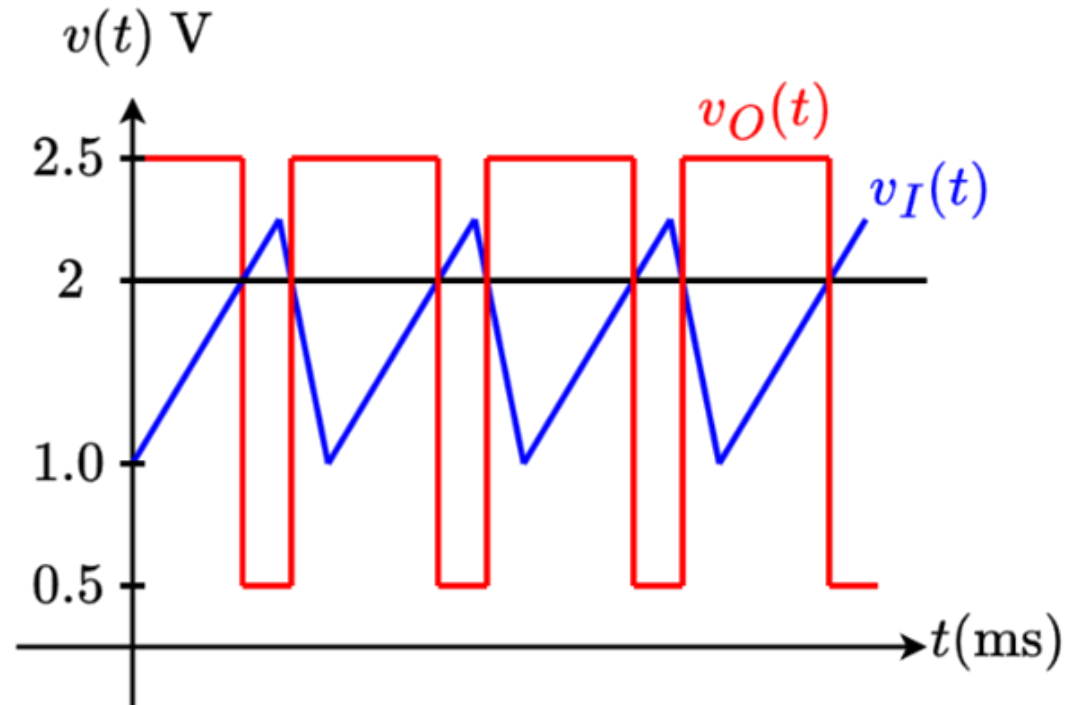


(b) Inverting: When E_i is above V_{ref} , $V_o = -V_{sat}$.



Open Loop Configuration: Example 2

- Design a circuit using OP Amp from the waveform graph.

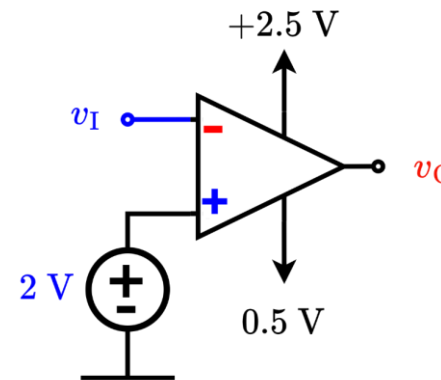


Solution:

v_I smaller than 2 V $\Rightarrow v_O = 2.5$ V $\rightarrow$ **Positive Saturation**

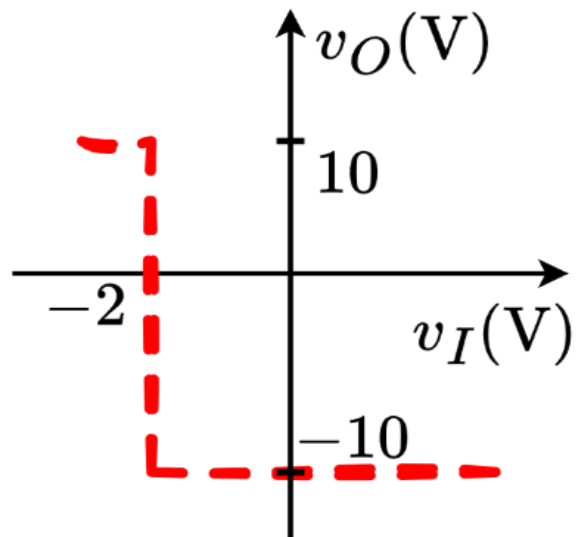
v_I larger than 2 V $\Rightarrow v_O = -2.5$ V $\rightarrow$ **Negative Saturation**

INVERTING CONFIGURATION



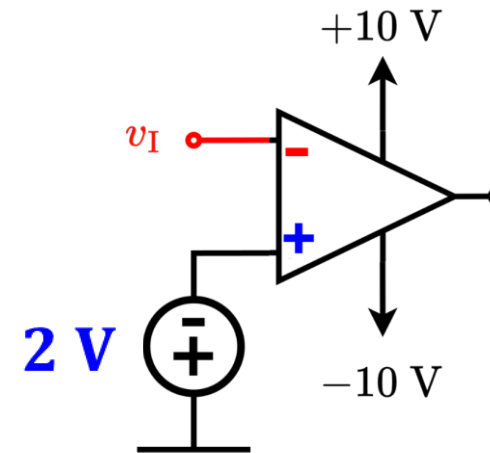
Open Loop Configuration: Example 1

- Design a circuit using **op-amp** that has the voltage transfer characteristics as shown in the figure below. $v_o(V)$ is the **output voltage** and $v_i(V)$ is the **input voltage**.

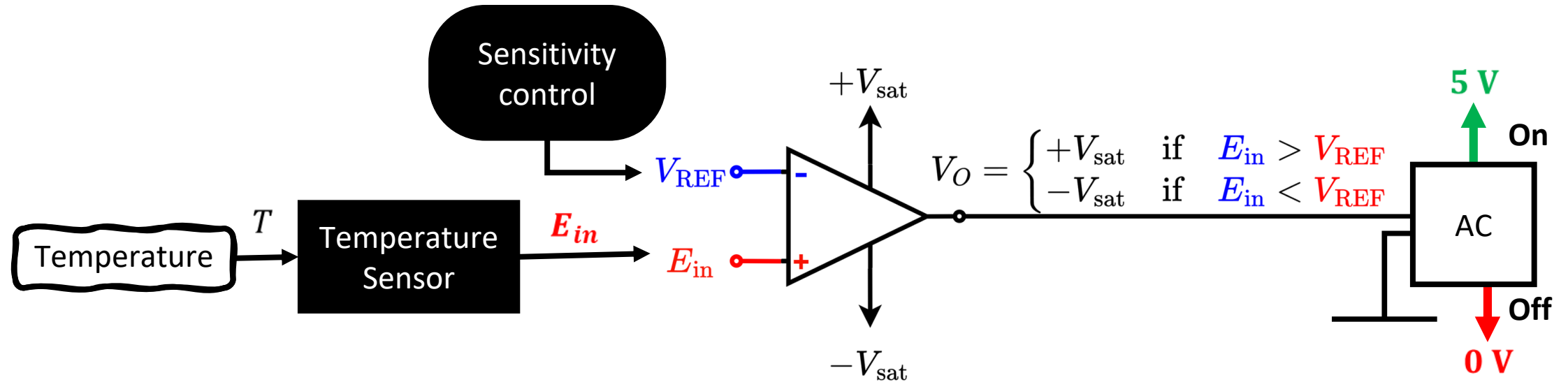


Solution:

Inverting comparator



Comparator Application – Automatic AC



- Comparators can be used to switch on an **AC** based on a **Temperature**.

Temp. $\uparrow$ **AC** - ON

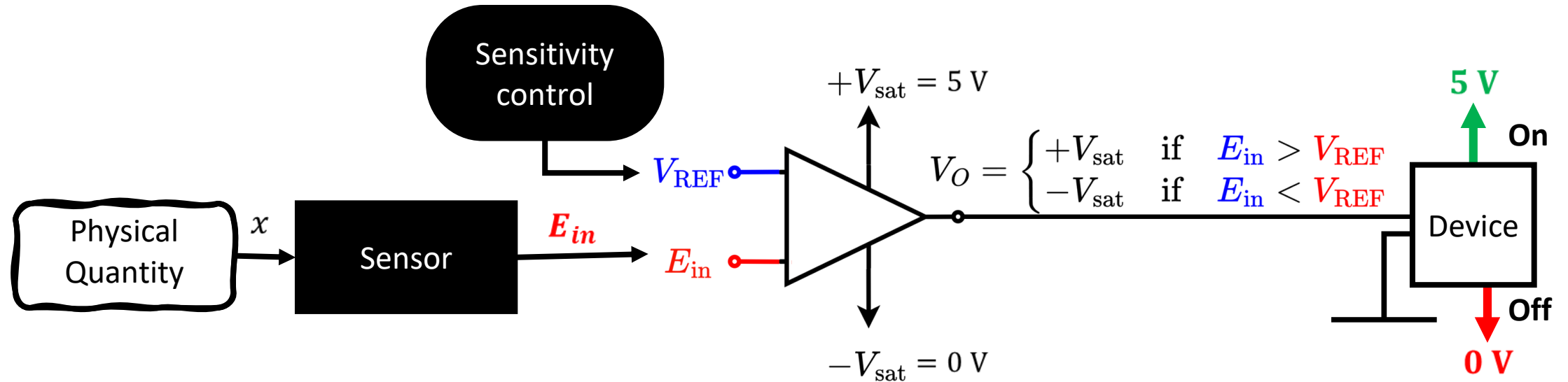
Temp. $\downarrow$ **AC** - OFF

Temp $\uparrow \propto E_{in} \uparrow \rightarrow \text{AC} - \text{ON}$ $\rightarrow$ NON-INVERTING RELATIONSHIP: **NON INVERTING**

COMPARATOR

- The sensitivity of switching on the **AC** is determined by V_{REF}

Comparator Application – Summary



- Comparators can be used to switch on **any device** based on a **Physical Quantity**.

Phys. Qnty. $\uparrow$ **DEVICE** – ON / OFF

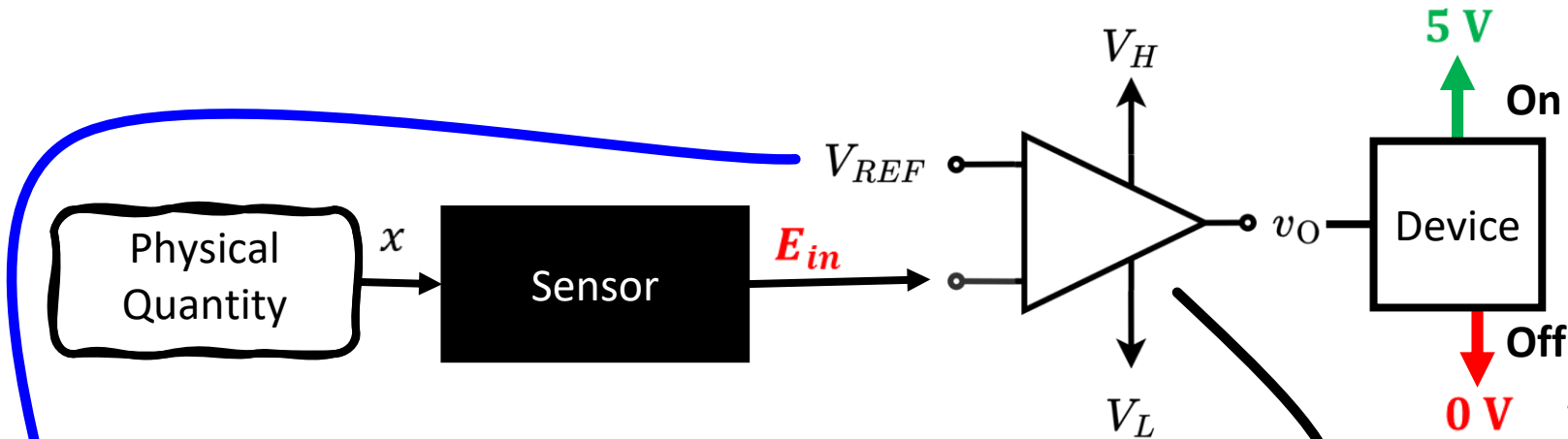
Phys. Qnty. $\downarrow$ **DEVICE** – OFF / ON

Phys. Qnty. $\rightarrow$ $E_{in} \downarrow \uparrow \rightarrow$ **DEVICE** – ON/OFF

This Relationship:
Determines the type of **Comparator**

- The sensitivity of **switching on the AC** is determined by V_{REF}

Solving Comparator Problems

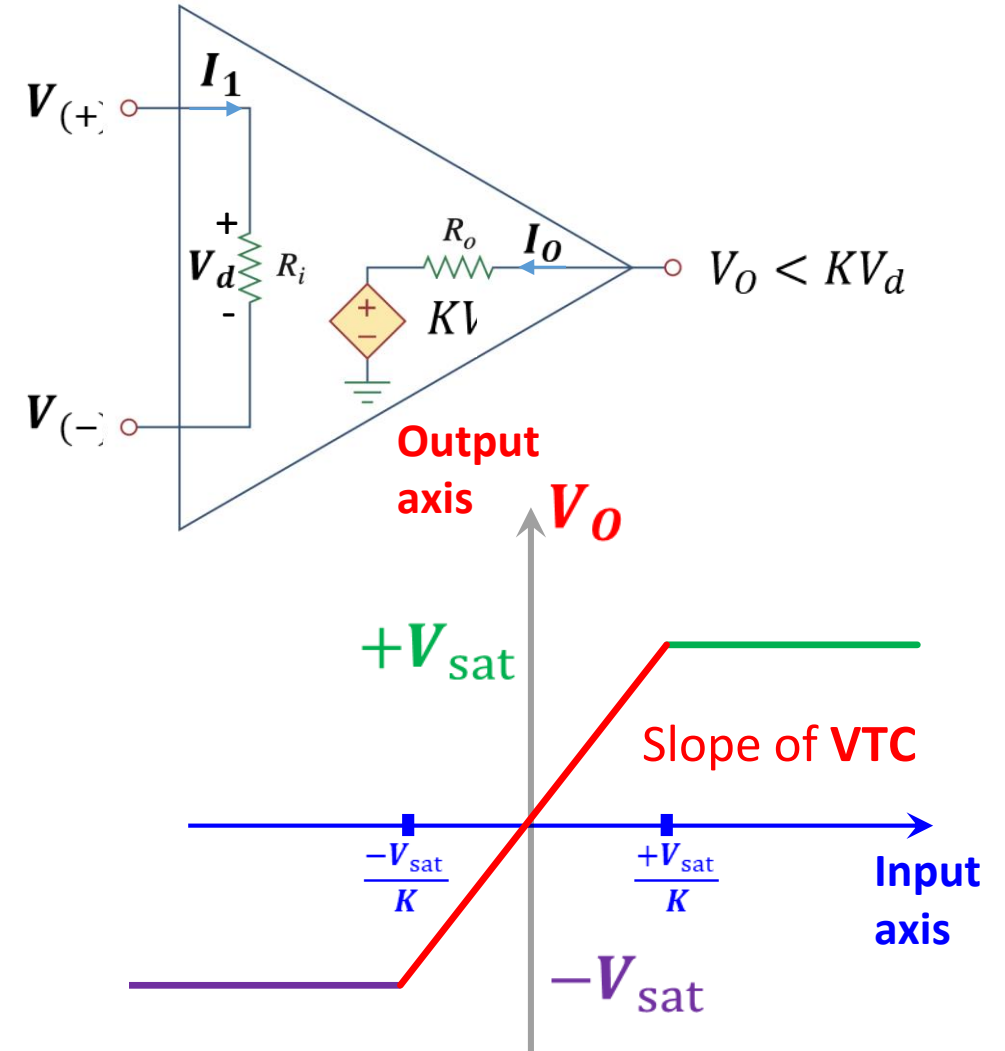


1. Determine type of comparator from the $E_{in} \rightarrow v_O$ relationship.
2. Determine V_{REF} from the switching condition from the provided information
3. Determine the saturation voltages V_H/V_L from the output device

Op-amp Model parameters

- Input resistance: R_i
- Output resistance: R_o
- Voltage (Differential/OL) Gain: A/A_{OL} or K
- Positive Saturation Voltage: $+V/V_{CC}/V_{sat}$
- Negative Saturation Voltage: $-V/V_{EE}/-V_{sat}$

Parameter	Typical Range	Ideally



Thank you!

Lecture 3 ends here